

The AI Agent Testing Pyramid

A Hands-On Workshop — four levels, four exercises, one robust testing strategy

WORKSHOP

AI AGENT TESTING



The Core Problem: AI Agents Are Non-Deterministic

Same input. Different output. Every time.

Traditional Testing: Deterministic

Same input, same output.

Pass or fail is clear.

30+ years of tooling built on this contract.

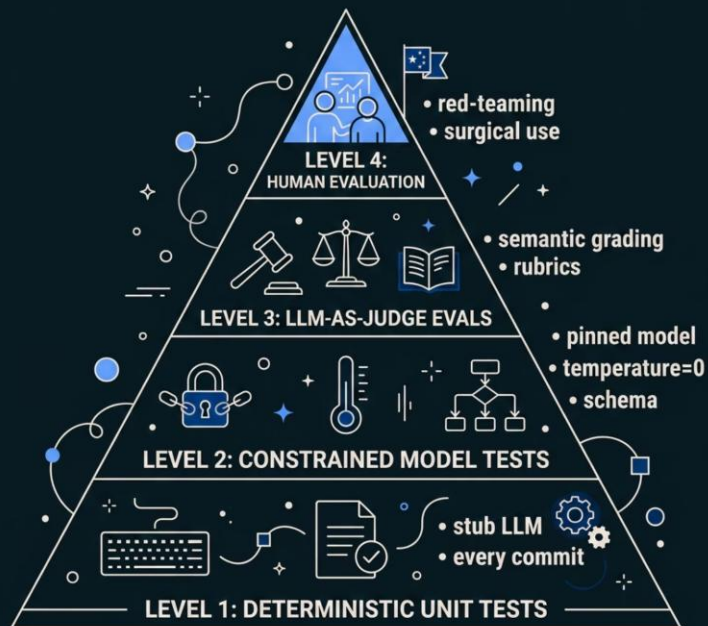
AI Agent Testing: Non-Deterministic

Same prompt, five different answers.

Wording, structure, and reasoning all vary.

Classic pass/fail breaks down.

The Full Pyramid



What each tier buys you

L1 — Speed

Milliseconds per test.
Catches broken plumbing instantly.

L2 — Semantics

Automates quality checks
humans can't scale.

L3 — Accuracy

Validates prompt contracts
before they reach
production.

L4 — Truth

Ground truth for what users
actually experience.

How the Exercises Work: "Flip the Prompt"



The Method

You set the context in **Claude Code** — then Claude interviews *you* to fill in the details.

→ Type the prompt

Paste the exercise starter prompt into Claude Code

→ Answer the questions

Claude asks; you respond with your agent's specifics

→ Get working code

Claude generates tests tailored to your context

EXERCISE 1

LEVEL 1 — BASE TIER

Deterministic Unit Tests

Verify the agent's deterministic plumbing — no LLM required.

What you test

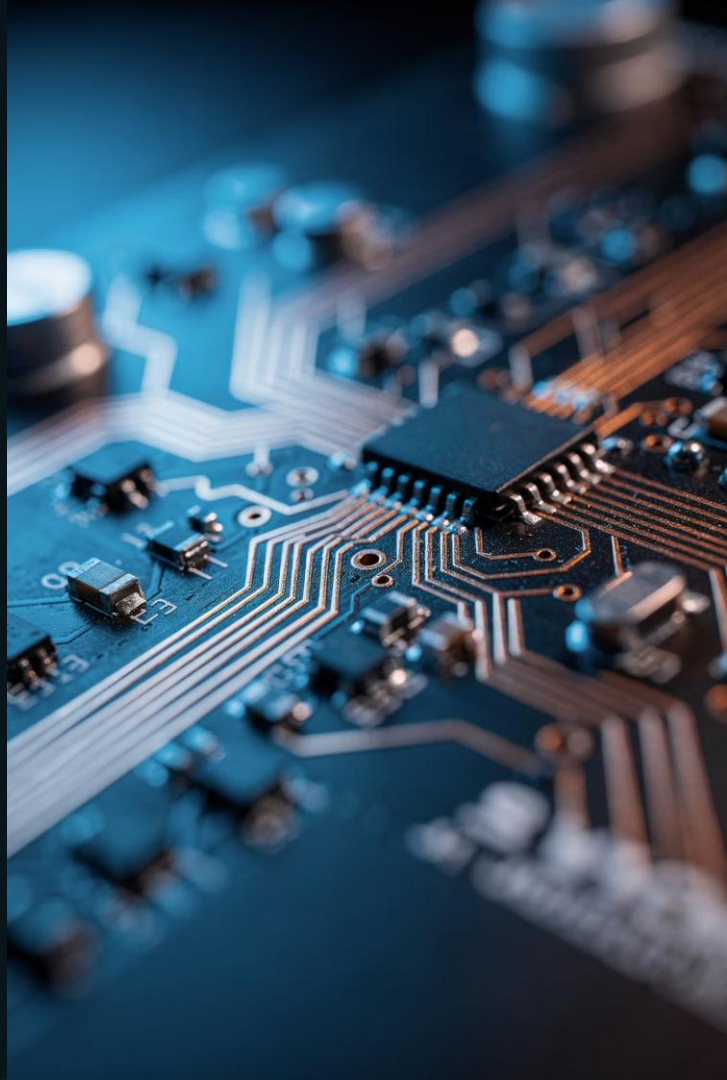
Orchestration logic, tool routing, error handling — with the LLM fully stubbed out

The seam

Establishes the `SearchTool Protocol` that makes every higher tier trustworthy

Speed

Fast, cheap, runs on **every commit** — maximum feedback loop velocity





EXERCISE 2

LEVEL 2 — LOWER-MIDDLE TIER

Constrained Model Tests

Real model calls — but tightly controlled inputs and outputs.

Pinned model

`claude-haiku-4-5` at
`temperature=0` for
reproducible results

Schema enforcement

JSON schema validation
ensures structured,
predictable outputs every
run

Prompt regressions

Catches prompt drift that a stubbed L1 test can never see



EXERCISE 3

LEVEL 3 — UPPER-MIDDLE TIER

LLM-as-Judge Evals

Use a second model to grade outputs against calibrated rubrics.

What it catches

Semantic correctness, tone drift, reasoning quality, hallucinations



The judge is itself a system under test — rubric calibration is essential.

Beyond structured checks

Finds regressions that schema validation completely misses

EXERCISE 4

LEVEL 4 — TOP TIER

Human Evaluation

The highest-signal layer — applied surgically to release candidates.

Preference testing

Side-by-side comparisons
revealing what users actually
prefer

Expert review

Domain specialists validate
accuracy, nuance, and
professional quality

Red-teaming

Adversarial probing for edge cases, safety issues, and failure modes

❏ Expensive and slow by design — reserve for what matters most.



Key Takeaways

Start at the base

L1 stubs give you fast, deterministic confidence on every commit

Move up deliberately

Each tier catches failures the tier below cannot detect

Reserve L4 for what counts

Human eval is your highest signal — spend it on release candidates

✓ You now have a repeatable framework for testing any AI agent — from unit to human scale.

